



Stellenbosch
UNIVERSITY
IYUNIVESITHI
UNIVERSITEIT

Portable Coordinate Measuring Arm Joint Design

Mechatronic Project 478
Final Report

Author: Lena Matota
24989142

Supervisor: Prof. K Schreve

2026/07/27

Department of Mechanical and Mechatronic Engineering
Stellenbosch University
Private Bag X1, Matieland 7602, South Africa.

Copyright © 2023 Stellenbosch University.
All rights reserved.

Plagiarism Declaration

1. Plagiarism is the use of ideas, material and other intellectual property of another's work and to present it as my own.
2. I agree that plagiarism is a punishable offence because it constitutes theft.
3. I also understand that direct translations are plagiarism.
4. Accordingly all quotations and contributions from any source whatsoever (including the internet) have been cited fully. I understand that the reproduction of text without quotation marks (even when the source is cited) is plagiarism.
5. I declare that the work contained in this assignment, except where otherwise stated, is my original work and that I have not previously (in its entirety or in part) submitted it for grading in this module/assignment or another module/assignment.

Student number	Signature
Initials and surname	Date

AI Use Declaration

1. I am convinced and can support my claim that my assessment product is an indication of my own learning, knowledge, skills, and understanding.
2. Where I have used AI tools for enhancing my own creation of ideas and words, I acknowledge that I have to declare it.
3. Where I have used AI tools for generating new ideas, words, code, image-prompts for other AI Image-generating tools, or structure and even presentations (or other AI tools that can be used as assistants to the knowledge building and representing process), I have declared and documented the use of such tools and I am prepared to talk about the process I used and what it contributed to my learning and insights.
4. I am aware that the lecturer can ask me to demonstrate my learning, for example through explaining the choices I made in terms of approach, content used, literature selected, conclusions drawn, etc. through an additional assessment like an oral (for example).
5. I understand that if I am not able to agree to the above points, there is a chance that my academic behaviour will be deemed unethical and might lead to a disciplinary case being brought against me on the grounds of cheating or plagiarism and that the standard procedures for such behaviour will be followed.
6. As per the Disciplinary Code of SU (par. 10.2.1 and 10.2.2), I understand that I take responsibility for the integrity of my work, which includes the obligation to ask for clarification from an academic member of staff if I am unsure of anything, and that I strictly adhered to all instructions received in the course of the academic assessment by relevant and authorized staff (whether the instruction is in oral or written format).
7. I understand that when I am not able to document and declare my use of AI tools, this behaviour will be deemed as cheating in examinations and assessments (Disciplinary Code 1.1 c.) as I referred to “unauthorized notes, books, electronic devices or other reference material”.

Signature:

Name: Student no:

Date:

Executive summary

Title of Project
Joint Design for a Portable Coordinate Measuring Arm (PCMA)
Objectives
To derive technical performance measures (TPMs) from an error-propagation model, and use them to design, build, and test a position-holding joint that maintains and /or improves the arm's overall accuracy.
What is current practice and what are its limitations?
Current joints rely on counterbalancing and lack holding torque, meaning the operator must physically support the arm's weight during operation.
What is new in this project?
The introduction of a position-holding mechanism specifically engineered using mathematically derived TPMs.
If the project is successful, how will it make a difference?
It will directly maintain or improve the physical measurement accuracy of the arm. Additionally, it will yield a modified kinematic model to serve as a generalized framework for future accuracy analysis.
What are the risks to the project being a success? Why is it expected to be successful?
Risks: University workshop manufacturing capabilities may fall short of required tolerances, and the holding mechanism could introduce unwanted friction. Expectation of Success: The design requirements are heavily grounded in an established, proven error model.
What contributions have/will other students made/make?
The project builds directly upon an initial error-propagation model developed by Stellenbosch alumna Leon Hall.
Which aspects of the project will carry on after completion and why?
The modified error-propagation model. It will be able to accept custom arm dimensions and compute the uncertainty distribution within a given work volume, forming the mathematical foundation for future PCMA development.
What arrangements have been/will be made to expedite continuation?

Extensive, standardized documentation will be left behind to ensure future researchers can build directly onto this work without repeating groundwork.

ECSA Exit Level Outcomes

This section addresses the following Engineering Council of South Africa (ECSA) exit level outcomes and indicates how each outcome was demonstrated in the project.

GA 1 - Problem-Solving

The project addresses an engineering problem, which is the lack of a joint design for PCMA's that meet accuracy requirements that account for spatial dependency and proper position-holding implementation without the impedance of uninhibited movement.

GA 2 - Application Scientific and Engineering Knowledge

GA 3 - Engineering Design

GA 5 - Use of Engineering Tools

GA 6 - Professional and Technical Communication

GA 8 - Individual and Collaborative Teamwork

GA 9 - Independent Learning Ability

Acknowledgements

Table of contents

Executive summary	iv
ECSA Exit Level Outcomes	vi
List of figures	x
List of tables	xi
List of symbols	xii
1 Introduction	1
1.1 Background	1
1.2 Aim and Objectives	2
1.3 Scope	2
1.4 Motivation	2
1.5 Use of AI Tools	3
1.6 Report Overview	3
2 Literature review	4
2.1 PCMA Context	4
2.2 The Denavit-Hartenberg Model	4
3 Requirement Analysis	5
3.1 Identification of Stakeholders	5
3.2 System Context	6
3.3 Design Independent Parameters	6
3.4 Functional Requirements	7
3.5 Technical Performance Measures	7
4 Concept Development	9
4.1 Concept Generation	9
4.1.1 Concept One: PCMA Joint with a Locking Dial	9
4.1.2 Concept Two: PCMA Joint with a Counterbalance Spring	10
4.1.3 Concept Three: PCMA Joint with an Electromagnetic Brake	10
4.2 Concept Evaluation	11

4.3	Concept Selection	14
4.4	Design Refinement	14
5	Design Analysis	15
5.1	Mechanical Design	15
5.2	Electronic Design	15
5.3	Software Design	15
6	Design Evaluation	16
6.1	As-Built System Overview	16
6.2	Performance Testing	16
7	Experimental Procedure	17
7.1	Experimental Apparatus	17
7.2	Experimental Setup	17
7.3	Experimental Equipment Specifications	17
7.3.1	Test Sphere Specifications	17
7.3.2	Joint Component Specifications	17
7.3.3	Experimental Apparatus Specifications	17
7.3.4	Methodology	17
8	Conclusions	18
A	Mathematical proofs	19
A.1	Euler's equation	19
A.2	Navier Stokes equation	19
B	Experimental results	20
	List of references	21

List of figures

3.1	System Context	6
3.2	Top Level Functional Decomposition	7
4.1	PCMA Joint with Locking Dial	10
4.2	PCMA Joint with Couunterbalance Spring	10
4.3	PCMA Joint with Electromagnetic Brake	11

List of tables

3.1	Identified stakeholders and their classifications	5
3.2	Design Independent Parameters	6
3.3	Table 3: Functional Requirements	7
3.4	Technical Performance Measures	8
4.1	Concept Evaluation	12

List of symbols

Constants

$$L_0 = 300 \text{ mm}$$

Variables

Re_D	Reynolds number (diameter)	[]
x	Coordinate	[m]
$\ddot{x}$	Acceleration	[m/s ²]
θ	Rotation angle	[rad]
τ	Moment	[N·m]

Vectors and Tensors

$\vec{v}$ Physical vector, see equation ...

Subscripts

a	Adiabatic
a	Coordinate

Abbreviations

3D	three-dimensional
AI	artificial intelligence
CAD	computer-aided design
D-H	Denavit-Hartenberg
DIP	design independent parameters
EC	evaluation criteria
FR	functional requirements
NSoI	narrower system of interest
PCMA	portable coordinate measuring arm
SPAT	single-point articulation test
TPM	technical performance measure
WSoI	wider system of interest

Chapter 1

Introduction

1.1 Background

A portable coordinate measuring arm (PCMA) is a handheld metrology instrument that captures three-dimensional (3D) coordinates directly from a physical object, as shown in 1.1 below, (1).



Figure 1.1: A portable coordinate measurement arm, (1)

PCMA's offer adaptable measurement systems for multiple applications, such as dimensional analysis, computer-aided design (CAD) inspection, reverse engineering, etc., (2). The main limitation of PCMA's is the effect of manual operation, which can introduce positional instability and dynamic structural deformation during measurement, (3). Manual operation can also inhibit calibration procedures as it is required that the PCMA must be held still during the procedure, (4).

As a result, designing an effective joint with a position-holding mechanism would improve the design of a PCMA and a good practice of safety. By designing the joint in this manner, error due to manual operation will be minimised, kinematic stability and accuracy will be improved.

1.2 Aim and Objectives

The aim of this project is to implement a position-holding mechanism on a PCMA without jeopardising its accuracy and ease of movement.

The objectives of the project are:

1. To generalise the conventional Denavit-Hartenberg(D-H) model derived by Hall and Schreve by sampling the working volume of a PCMA determining distribution of uncertainty along its length, (5).
2. To design, build, and test the joint design through the single-point articulation test (SPAT), such that it maintains and/or improves the accuracy of the PCMA.
3. To compare the results from SPAT to the D-H model and verify the joint design's effect on the PCMA's accuracy.

1.3 Scope

In the process of determining D-H parameters, operation manual and available schematics will be used. In addition, only joint will be designed, built, and tested for data acquisition, through calibration and accuracy testing via SPAT.

1.4 Motivation

PCMA's are becoming more popular than tradition coordinate measuring machines (CMM) for in-line¹ and on-site industrial measurement, (1). They have broad applicability across multiple industries such as aerospace, automotive, heavy manufacturing, where having a portable measurement instrument decreases handling time, production delays, and revision of work.

Despite their growing popularity, the PCMA's revolute joints must satisfy competing demands, including:

- Tight run-out tolerances for spatial accuracy.
- Sufficient holding torque to keep the arm stationary once released.
- Internal cable routing.

This has proved difficult since there are found commercial available joints that meet these demands. In addition, it has been established that angular position errors at each joint propagate through the kinematic chain, which affects end-effector's

¹In the context of PCMA's, "in-line" refers to performing quality inspections and measurements directly on the production shop floor, instead of a temperature-controlled quality laboratory, (1).

accuracy, proving that joints have a measurable impact on the PCMA's accuracy, (6). This project addresses the gap by deriving a joint specifications from a generalised model based off of an error model that was developed by a Stellenbosch Alumnus, Leon Hall, (5). This will produce a validated joint and testing methodology to support future PCMA development.

1.5 Use of AI Tools

Gemini is an artificial intelligence (AI) tool will be used check grammatical errors and sentence structure as well as LaTeX formatting for the report.

1.6 Report Overview

This section follows the following structure:

- Section 1 introduces the project background, aim and objectives, scope, and motivation for project.
- Section 2 presents a literature review discussing sources relevant to PCMA joint design, position-holding technology, kinematic modeling, error analysis, and PCMA testing.
- Section 3 details the PCMA's requirement analysis.

Chapter 2

Literature review

2.1 PCMA Context

Sources such as Leussink Engineering and Meagher gave context as what PCMA's are and how they are used.

2.2 The Denavit-Hartenberg Model

Chapter 3

Requirement Analysis

The objective of the requirement analysis is to establish clear expectations of the PCMA's joint design for the purposes of concept development.

3.1 Identification of Stakeholders

The first step in the requirement analysis process is to identify stakeholders from the operational and support life cycle phases of the project. Stakeholders will be classified into the following categories, as defined below:

- Primary stakeholders directly interact with the system while it is in operation.
- Secondary stakeholders contribute to the development and functioning of the system.
- Indirect stakeholders rely on or are affected by the system, but do not interact with the system directly.

The stakeholders relevant to the project are defined in Table 3.1 below.

Table 3.1: Identified stakeholders and their classifications

Stakeholder ID	Description	Classification
SID_01	Supervisor of the project	Primary
SID_02	Students researching PCMA development	Secondary
SID_03	Companies invested in PCMA development	Indirect
SID_04	Mechanical workshop staff	Indirect
SID_05	Mechatronics laboratory managers	Indirect

3.2 System Context

The next step is to define the narrower system of interest (NSoI) and the wider system of interest (WSoI) of the project¹, (7) for the purposes of this project, the joint was identified as the NSoI and WSoI is the NSoI with a reliable power supply and an artefact with which the joint will be tested. Figure 3.1 illustrates the system context, with the NSoI shown as a "black box", (7).

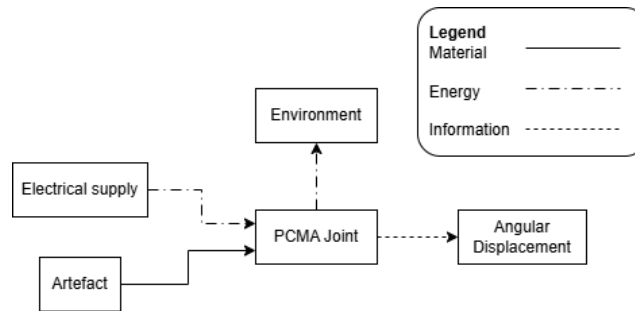


Figure 3.1: System Context

3.3 Design Independent Parameters

Next, the design independent parameters (DIPs) are defined. DIPs influence the system during the operational and support life cycle stages, but are not controllable nor are part of the system being defined, (7). The DIPs of the joint design are detailed in Table 3.2 below.

Table 3.2: Design Independent Parameters

DIP_ID	Description	Expected range
DIP_01.1	Electrical supply	12 V - 24 V
DIP_01.2	Electrical supply	12 A maximum
DIP_01.3	Electrical supply	Direct current
DIP_02.1	Artefact	Compliant with ISO 10360-12:2016
DIP_02.2	Artefact	20 points reference points
DIP_03.1	Environment	Laboratory ambient temperature: 0 °C - 35 °C
DIP_03.2	Environment	Laboratory ambient humidity: 20 % - 80 %
DIP_03.3	Project budget	R 6000

¹The NSoI is the system being designed and the WSoI includes the NSoI and elements that the NSoI interacts with, (7).

3.4 Functional Requirements

In this section, the primary and top-level derived functional requirements (FRs) for the joint design are defined. The only FRs that are considered are only associated with joint's operation. FRs are categorised into primary and derived FRs, in which:

- Stakeholders are directly concerned with primary FRs.
- Derived FRs are required so that the system can perform primary FRs.

As seen in Table 3.3 below, the FRs for the joint design are listed. In Figure 3.2,

Table 3.3: Table 3: Functional Requirements

FR ID	Description	Primary / Derived
FR_01	Articulate output link	P
FR_02	Hold position	P
FR_03	Measure angular displacement	P
FR_04	Store angular displacement data	D
FR_05	Release position	P

the functional requirements of the joint are depicted below.

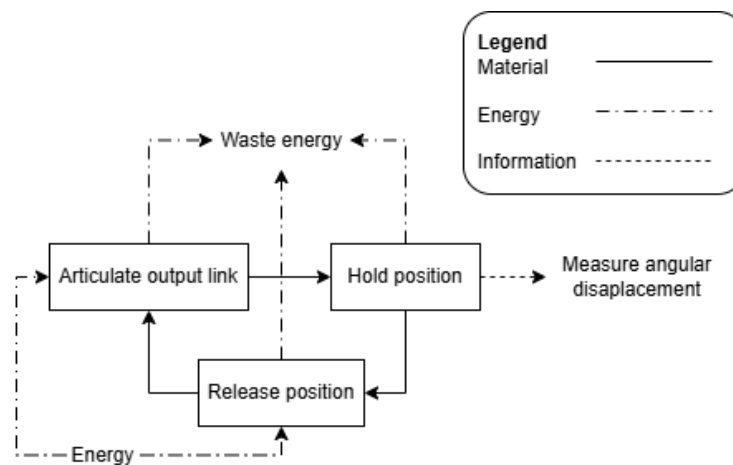


Figure 3.2: Top Level Functional Decomposition

3.5 Technical Performance Measures

This section defines the technical performance measures (TPMs) which will be used to determine how well the joint design (NSoI) performs the required

functions². The TPMs, listed in Table 3.4, were derived from the results obtained from the generalised error-propagation model.

Table 3.4: Technical Performance Measures

TPM ID	Related FR	Description	Target range
TPM_01	FR_01, FR_05	Operational articulation angle	90°
TPM_02	FR_01, FR_05	Run-out tolerance	1.571×10^{-3} rad
TPM_03	FR_02	Holding torque	7.725 Nm
TPM_04	FR_03, FR_04	Encoder accuracy	1000 PPR
TPM_05	FR_01, FR_02, FR_05	Joint life cycle	10^6 life cycles

²TPMs are measurable and unambiguous parameters that can be influenced by designing the joint

Chapter 4

Concept Development

4.1 Concept Generation

The generalised error propagation model was used to derive the TPMs in the development of the joint design. These TPMs guided the research into various mechanical solutions, and different mechanisms were evaluated for implementation to ensure the final design could achieve the required TPMs. Three concepts were generated and evaluated against the FRs and the TPMs, as presented in this section.

4.1.1 Concept One: PCMA Joint with a Locking Dial

Figure 4.1 below shows a PCMA joint with the following internal modifications to implement position-holding capabilities. In the center of the shaft, is a spacer between angular contact bearings, which will allow the shaft to rotate with minimal run-out. Bellville washer and preload nuts are used to secure the bearings in place. On one end, the rotary encoder's shaft is connected to the joint's shaft via a rigid coupling and the encoder's body is held secure by the joint housing. This is done to ensure that the relative angular displacement of the shaft with respect to the joint housing is measured. The joint housing is held stationary by the base link and the joint shaft is rotated by the output link. The knob on the other end of the shaft is meant to loosen and/or tighten the preload nuts against the bearings to lock or release the shaft for the output link's rotation.

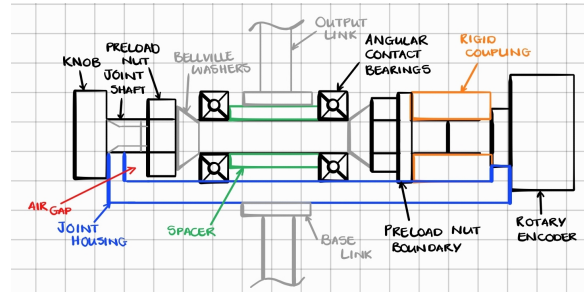


Figure 4.1: PCMA Joint with Locking Dial

4.1.2 Concept Two: PCMA Joint with a Counterbalance Spring

Figure 4.2 illustrates a PCMA joint with external modifications to implement position-holding. The counterbalance spring produces a lifting force to the upper link via the upper and lower rods. Each rod is pin-connected to a cuff that can be fixed in position along the length of either link using a set screw, thereby holding the position of the joint. The

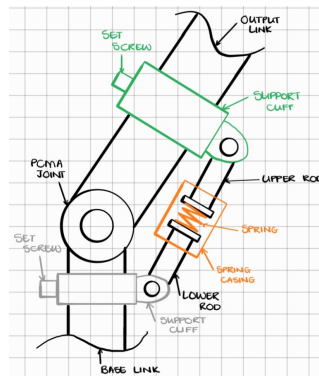


Figure 4.2: PCMA Joint with Counterbalance Spring

4.1.3 Concept Three: PCMA Joint with an Electromagnetic Brake

Figure 4.3 depicts a PCMA with internal modifications to implement an electromagnetic brake. In the center of the shaft, two angular contact bearings are fitted in place with the bearing housing and pressed against a shoulder of the shaft by preload nuts and washers. On one side of the shaft, the rotary encoder's shaft is connected to the joint's shaft via a rigid coupling and the encoder's body is held stationary by the joint housing. The base link supports the joint housing. The electromagnetic brake sits on the other side of the shaft and is also held stationary by the joint housing, however, the brake disc is free to rotate on the joint shaft. When the brake is energized, the air gap between the brake disc and friction plate opens up and the shaft is free to be rotated by the output link.

When the brake's power supply is cut off, the air gap closes and the joint is held still.

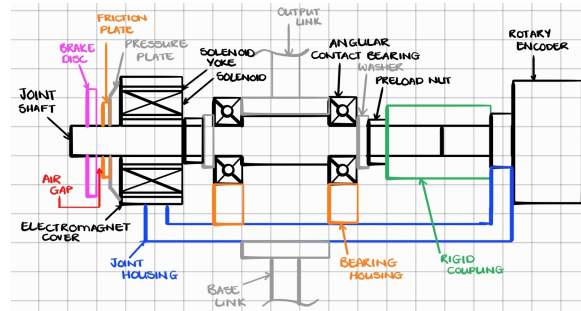


Figure 4.3: PCMA Joint with Electromagnetic Brake

4.2 Concept Evaluation

The concepts presented in the previous section were determined to be feasible. A qualitative evaluation of the concepts is to be conducted as the generated concepts are still solution neutral and cannot undergo a full quantitative comparison. This section compares the concepts in terms of the following evaluation criteria (EC):

- EC1 Load capacity: The ability of the joint design to carry the required holding torque (TPM_03).
- EC2 Operational articulation angle: The ability of the joint design to achieve the full required sweep without interference (TPM_01).
- EC3 Positional stability: The ability of the joint design to maintain position when the position-holding technology is in use (TPM_02).
- EC4 Component availability: The degree to which all required components are available in the mechanical workshop.
- EC5 Construction simplicity: The number of distinct components and the ease of assembly and alignment during fabrication.

EC1 was derived from TPM_03 and determined to be the most important EC as the joint will be ineffective if it is unable to achieve the holding torque dictated by TPM_03. EC2 was derived from TPM_01 as the chosen joint design must not inhibit the PCMA's existing range of motion once implemented. EC3 was derived from TPM_02 as any internal sources of error must be minimised to maximise the joint design's ability to maintain and/or improve the PCMA's accuracy. EC4 and EC5 was included as a practical design consideration given the manufacturing capabilities of the mechanical workshop and the project's timeline and budget.

Table ?? below, presents a comparison of the generated concepts. To rate the concepts, one of them is marked as the DATUM and the other two are awarded a +1 if the other concept is stronger or –1 if it is weaker. Each rating is motivated in the table as well.

Table 4.1: Concept Evaluation

Criterion		Concept 1: PCMA Joint with a Locking Dial			Concept 2: PCMA Joint with a Counterbalance Spring (DATUM)		Concept 3: PCMA Joint with an Electromagnetic Brake	
#	Description	Wt.	Rtg	Motivation	Dat	Motivation	Rtg	Motivation
EC1	Load capacity	5	+1	A GT2 timing belt driven by a stepper motor achieves positioning equivalent to the datum at the 1–2 mm target. FR ₁₃ already mandates a real-time encoder for user-interface state reporting; with encoder feedback available to all concepts equally, residual belt stretch is corrected in closed loop, removing any inherent accuracy disadvantage.		Datum: PWM-controlled DC motor with piston; accuracy is limited by gear backlash without a feedback sensor.	0	The lead screw's open-loop step resolution is high, but since FR ₁₃ already requires an encoder for real-time user-interface feedback, this inherent advantage is negated when compared with an encoder-equipped belt system at the 1–2 mm target range.
EC2	Operational articulation angle	5	+1	A GT2 timing belt driven by a stepper motor achieves positioning equivalent to the datum at the 1–2 mm target. FR ₁₃ already mandates a real-time encoder for user-interface state reporting; with encoder feedback available to all concepts equally, residual belt stretch is corrected in closed loop, removing any inherent accuracy disadvantage.		Datum: PWM-controlled DC motor with piston; accuracy is limited by gear backlash without a feedback sensor.	0	The lead screw's open-loop step resolution is high, but since FR ₁₃ already requires an encoder for real-time user-interface feedback, this inherent advantage is negated when compared with an encoder-equipped belt system at the 1–2 mm target range.

Table 4.1 – Continued from previous page

Criterion			Concept 1: PCMA Joint with a Locking Dial		Concept 2: PCMA Joint with a Counterbalance Spring (DATUM)		Concept 3: PCMA Joint with an Electromagnetic Brake	
#	Description	Wt.	Rtg	Motivation	Dat	Motivation	Rtg	Motivation
EC3	Positional stability	5	+1	A GT2 timing belt driven by a stepper motor achieves positioning equivalent to the datum at the 1–2 mm target. FR_13 already mandates a real-time encoder for user-interface state reporting; with encoder feedback available to all concepts equally, residual belt stretch is corrected in closed loop, removing any inherent accuracy disadvantage.		Datum: PWM-controlled DC motor with piston; accuracy is limited by gear backlash without a feedback sensor.	0	The lead screw's open-loop step resolution is high, but since FR_13 already requires an encoder for real-time user-interface feedback, this inherent advantage is negated when compared with an encoder-equipped belt system at the 1–2 mm target range.
EC4	Component availability	5	+1	A GT2 timing belt driven by a stepper motor achieves positioning equivalent to the datum at the 1–2 mm target. FR_13 already mandates a real-time encoder for user-interface state reporting; with encoder feedback available to all concepts equally, residual belt stretch is corrected in closed loop, removing any inherent accuracy disadvantage.		Datum: PWM-controlled DC motor with piston; accuracy is limited by gear backlash without a feedback sensor.	0	The lead screw's open-loop step resolution is high, but since FR_13 already requires an encoder for real-time user-interface feedback, this inherent advantage is negated when compared with an encoder-equipped belt system at the 1–2 mm target range.
EC5	Construction simplicity	5	+1	A GT2 timing belt driven by a stepper motor achieves positioning equivalent to the datum at the 1–2 mm target. FR_13 already mandates a real-time encoder for user-interface state reporting; with encoder feedback available to all concepts equally, residual belt stretch is corrected in closed loop, removing any inherent accuracy disadvantage.		Datum: PWM-controlled DC motor with piston; accuracy is limited by gear backlash without a feedback sensor.	0	The lead screw's open-loop step resolution is high, but since FR_13 already requires an encoder for real-time user-interface feedback, this inherent advantage is negated when compared with an encoder-equipped belt system at the 1–2 mm target range.
Number of + ratings			2				1	
Number of - ratings			1				1	

Table 4.1 – Continued from previous page

Criterion			Concept 1: PCMA Joint with a Locking Dial		Concept 2: PCMA Joint with a Counterbalance Spring (DATUM)		Concept 3: PCMA Joint with an Electromagnetic Brake	
#	Description	Wt.	Rtg	Motivation	Dat	Motivation	Rtg	Motivation
Overall total			+1				0	
Weighted total			+4		0		+1	

4.3 Concept Selection

4.4 Design Refinement

Chapter 5

Design Analysis

5.1 Mechanical Design

5.2 Electronic Design

5.3 Software Design

Chapter 6

Design Evaluation

6.1 As-Built System Overview

6.2 Performance Testing

Chapter 7

Experimental Procedure

7.1 Experimental Apparatus

7.2 Experimental Setup

7.3 Experimental Equipment Specifications

7.3.1 Test Sphere Specifications

7.3.2 Joint Component Specifications

7.3.3 Experimental Apparatus Specifications

7.3.4 Methodology

Chapter 8

Conclusions

Appendix A

Mathematical proofs

A.1 Euler's equation

Euler's equation gives the relationship between the trigonometric functions and the complex exponential function.

$$e^{i\theta} = \cos \theta + i \sin \theta \quad (\text{A.1})$$

Inserting $\theta = \pi$ in (A.1) results in Euler's identity

$$e^{i\pi} + 1 = 0 \quad (\text{A.2})$$

A.2 Navier Stokes equation

The Navier–Stokes equations mathematically express momentum balance and conservation of mass for Newtonian fluids. Navier-Stokes equations using tensor notation:

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_j} [\rho u_j] = 0 \quad (\text{A.3a})$$

$$\frac{\partial}{\partial t} (\rho u_i) + \frac{\partial}{\partial x_j} [\rho u_i u_j + p \delta_{ij} - \tau_{ji}] = 0, \quad i = 1, 2, 3 \quad (\text{A.3b})$$

$$\frac{\partial}{\partial t} (\rho e_0) + \frac{\partial}{\partial x_j} [\rho u_j e_0 + u_j p + q_j - u_i \tau_{ij}] = 0 \quad (\text{A.3c})$$

Appendix B

Experimental results

List of references

- [1] L. Engineering, "The advantages of using a portable cmm arm in metrology," *Leussink Engineering*, 2024.
- [2] S. Ibaraki and R. Saito, "Novel kinematic model of articulated arm coordinate measuring machine with angular position measurement errors of rotary axes," *CIRP Annals*, vol. 72, no. 1, pp. 449–452, 2023.
- [3] B. Alvarez, E. Cuesta, P. Luque, D. Mantaras, and D. Muina, "Dynamic deformations in coordinate measuring arms using virtual simulation," *International Journal of Simulation Modelling*, vol. 14, no. 4, pp. 609–620, 2015.
- [4] E. Cuesta, D. A. Mantaras, P. Luque, B. J. Alvarez, and D. Muina, "A calibration method of portable coordinate measuring arms by using artifacts," *MAPAN-Journal of Metrology Society of India*, vol. 34, no. 1, pp. 1–11, 2019.
- [5] L. Hall and K. Schreve, "Error distribution of an articulated arm measurement system," (Stellenbosch, South Africa), July 2010.
- [6] H. Meagher, "Measurement tool spotlight - portable measuring arms." OASIS Alignment Service, LLC, February 2016.
- [7] W. Callister and D. Rethwisch, *Material Science and Engineering*. Hoboken, NJ: Wiley, 8th ed., 2011.